A Project Report
Submitted in the partial fulfillment of the
requirements for the award of the degree of

BACHELOR OF TECHNOLOGY

In

DEPARTMENT OF COMPUTER SCIENCE ENGINEERING

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Declaration

The Project Report entitled **"Sports performance Analytics"** is a record of Bonafide work of teammembers**,K.Pavan–2420030112,B.Yugendra–2420030464,A.Sruti- 2420030640 , ,P.Bhavya-2420030506,**submitted in partial fulfilment for the award of B. Tech in Computer Engineering to K L University. The results embodied in this report have not been copied from any other departments/University/Institute.

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Certificate

This is to certify that the project-based report entitled **"Sports performance Analytics"** is a Bonafide work carried out and submitted by **K.Pavan–2420030112,B.Yugendra–2420030464,A.Sruti-2420030640 , ,P.Bhavya-2420030506**, in partial fulfilment of the requirements for the award of the degree of **Bachelor of Technology in Computer Science Engineering**, at **K L (Deemed to be University)**, during the academic year **2025–2026**.

Signature of the Supervisor

Signature of the HOD

Signature of the External Examiner

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ABSTRACT

Sports Performance Analytics is a web-based data analytics project developed to analyze and visualize player and team performance using interactive dashboards. The system helps coaches, analysts, sports organizations, and enthusiasts understand performance trends through statistical analysis and graphical representations. It processes sports-related datasets to calculate important performance metrics and present meaningful insights using modern visualization techniques. By transforming raw data into easy-to-understand charts and reports, the application simplifies performance evaluation and supports better decision-making.

The application is built using **React.js** for the frontend, providing a responsive and user-friendly interface for accessing sports analytics. It integrates data visualization libraries to generate bar charts, pie charts, line graphs, and comparative dashboards that help users explore player statistics and team performance. Users can compare multiple players, analyze historical records, identify strengths and weaknesses, and monitor improvements over time through interactive visualizations. The intuitive design ensures that even users with limited technical knowledge can easily interpret complex sports data.

The system analyzes various performance indicators such as matches played, runs scored, goals, assists, strike rates, averages, wickets, and other sport-specific statistics. These metrics are processed to identify performance trends, consistency levels, and overall contributions of players and teams. The dashboard provides statistical summaries and comparative analysis that help coaches formulate strategies, select players based on performance, and evaluate team effectiveness. The project demonstrates how data-driven analysis can enhance performance evaluation in modern sports.

Overall, Sports Performance Analytics highlights the importance of data analytics in improving sports performance and strategic planning. The project provides an efficient platform for analyzing historical sports data while presenting the results in a visually appealing and interactive manner. Its scalable architecture allows future enhancements such as real-time match analysis, integration with live sports APIs, predictive analytics using machine learning, and personalized performance recommendations. The project demonstrates how technology and analytics can be combined to improve decision-making, optimize player development, and contribute to the success of teams and athletes.

INTRODUCTION

In modern sports, data analytics has become an essential tool for improving player performance and team strategies. Large amounts of match and player data are generated during every game, making manual analysis difficult and time-consuming. Sports Performance Analytics provides an efficient platform to analyze this data and transform it into useful information.

The system allows users to visualize player statistics, compare different players, and monitor overall team performance using interactive charts and dashboards. By leveraging data analytics techniques, the project enables coaches and analysts to make informed decisions based on objective performance indicators.

The application is designed to provide a user-friendly interface where users can explore various performance metrics without requiring technical expertise.

Problem Description and Objectives:

Traditional methods of analyzing sports performance rely heavily on manual observation and basic statistics, which are often insufficient for understanding detailed player performance. As the amount of sports data continues to increase, extracting meaningful insights becomes more challenging.
The Sports Performance Analytics system addresses these challenges by automating data analysis, generating visual reports, and presenting performance trends through interactive dashboards.

LITERATURE SURVEY

Sports analytics has become an important part of modern sports due to advancements in data science and machine learning. Researchers use statistical analysis and predictive techniques to evaluate player performance, improve team strategies, and predict match outcomes. These methods help coaches make informed decisions based on performance data rather than intuition.

Data visualization plays a key role in sports analytics by presenting complex information through charts, graphs, and dashboards. Interactive visualizations make it easier to compare player statistics, identify trends, and monitor performance over different matches and seasons. This improves the efficiency of performance analysis.

Machine learning techniques are also widely used to predict player performance, detect injury risks, and recommend team selections. Algorithms analyze historical data to identify patterns and generate useful insights that support training and strategic planning. These technologies improve the accuracy and reliability of sports analysis.

Web-based sports analytics platforms provide users with easy access to performance data from anywhere. Technologies such as React.js and modern visualization libraries enable responsive dashboards that simplify data analysis. These platforms help coaches, analysts, and sports enthusiasts make better decisions through clear and interactive performance reports.

HARDWARE AND SOFTWARE REQUIREMENTS:

Hardware Specifications

The Sports Performance Analytics project is a lightweight web application that does not require high-end hardware. Since the application runs in a web browser, a standard computer with moderate specifications is sufficient for development and execution.

Hardware Specifications

Processor

- Intel Core i3 or equivalent processor
- Intel Core i5 or higher recommended for better performance

RAM

- Minimum: 4 GB
- Recommended: 8 GB

Storage

- Minimum 500 MB of free disk space
- Additional space for project files and dependencies

Display

- Standard monitor with a resolution of 1366 × 768 or higher

Internet Connection

- Required for downloading packages, deployment, and accessing online resources

The project is developed using modern web development technologies to provide a responsive and interactive user experience.

Operating System

- Windows 10/11
- Linux
- macOS

Programming Languages

- JavaScript (ES6+)
- HTML5
- CSS3

Frontend Technologies

- React.js
- Vite

Libraries

- Recharts (Data Visualization)
- Framer Motion (Animations)
- Lucide React (Icons)

Development Tools

- Visual Studio Code
- Node.js
- npm (Node Package Manager)

Deployment Platform

- Vercel

Web Browsers

- Google Chrome
- Microsoft Edge

IMPLEMENTATION

The Sports Performance Analytics system is implemented as a responsive web application using React.js and Vite. The application is designed with reusable React components, allowing efficient navigation between different sections while maintaining a fast and smooth user experience. The interface is responsive and adapts to various screen sizes, making it accessible on desktops, tablets, and mobile devices.

The project processes sports performance data and presents it through interactive dashboards. Performance statistics such as player achievements, team performance, match records, and other key metrics are organized and displayed using Recharts, which generates dynamic bar charts, pie charts, and line graphs. These visualizations help users compare performances and understand trends effectively.

To enhance the user experience, the application incorporates Framer Motion for smooth animations and transitions, creating an engaging interface. Lucide React is used to provide modern icons that improve navigation and the overall appearance of the application. The modular structure of React components makes the system easy to maintain, update, and extend with additional features in the future.

The project is developed using Visual Studio Code, with Node.js and npm managing dependencies and project execution. The final application is deployed on Vercel, allowing users to access the Sports Performance Analytics platform online through any modern web browser. The implementation demonstrates how modern frontend technologies can be combined to build an efficient, scalable, and user-friendly sports analytics application.

EXPEREMENTATION AND CODE:

Experimentation

The Sports Performance Analytics project was developed and tested to ensure that it accurately displays sports statistics and provides meaningful visual insights through an interactive web interface. The application was executed in a local development environment using Node.js and later deployed on Vercel for online accessibility. Different datasets were tested to verify the correctness of performance metrics, chart generation, and responsive design.

Steps Performed

1. Created the project using React.js and Vite.
2. Designed responsive user interface components for the dashboard.
3. Collected and organized sports performance data.
4. Processed the dataset and displayed player and team statistics.
5. Generated interactive bar charts, pie charts, and line graphs using Recharts.
6. Tested navigation, animations, and responsiveness on different devices and browsers.
7. Deployed the application on Vercel for public access.

src > App.jsx > ...

```
import React, { useState } from 'react';
import { AnimatePresence, motion } from 'framer-motion';
import { Shield, Cpu } from 'lucide-react';
import SportsGallery from './components/SportsGallery';
import PerformanceAnalysis from './components/PerformanceAnalysis';
import './App.css';

// 10 Sports Dynamic Metrics Database
const SPORT_METRICS_DATABASE = {
  Football: {
    overallScore: 94,
    recoveryScore: 88,
    fitnessLevel: 'Excellent',
    readinessStatus: 'Ready for full load',
    metrics: [
      { title: 'Performance Score', value: '94.2%', trend: '+1.4%', trendType: 'up' },
      { title: 'Speed', value: '34.2 km/h', trend: '+0.8', trendType: 'up' },
      { title: 'Acceleration', value: '6.4 m/s²', trend: 'Steady', trendType: 'neutral' },
      { title: 'Endurance', value: '92 / 100', trend: '+2.0', trendType: 'up' },
      { title: 'Strength', value: '84 / 100', trend: '+1.5', trendType: 'up' },
      { title: 'Agility', value: '88 / 100', trend: '+3.1', trendType: 'up' },
      { title: 'Power', value: '380 W', trend: '+12.4%', trendType: 'up' },
      { title: 'Accuracy', value: '91.8%', trend: '+2.8%', trendType: 'up' },
      { title: 'Reaction Time', value: '140 ms', trend: '-8ms', trendType: 'up' },
      { title: 'Recovery Score', value: '88%', trend: 'Optimal', trendType: 'up' },
      { title: 'Injury Risk', value: '12%', trend: '-0.15 Risk', trendType: 'up' },
      { title: 'Training Load', value: '650 score', trend: 'Optimal', trendType: 'neutral' },
      { title: 'Calories Burned', value: '840 kcal', trend: 'Session', trendType: 'neutral' },
      { title: 'Distance Covered', value: '10.8 km', trend: '+0.5', trendType: 'up' },
      { title: 'Sprint Count', value: '28', trend: '+4', trendType: 'up' },
      { title: 'Active Time', value: '84 mins', trend: 'Session', trendType: 'neutral' }
    ],
    matchStats: [
      { label: 'Expected Goals (xG)', value: '2.42' },
      { label: 'Pass Accuracy', value: '88.5%' },
      { label: 'Successful Tackles', value: '14' }
    ]
  },
  Cricket: {
    overallScore: 86,
    recoveryScore: 82,
    fitnessLevel: 'Optimal',
    readinessStatus: 'Ready for full load',
    metrics: [
      { title: 'Performance Score', value: '86.5%', trend: '+0.5%', trendType: 'up' },
      { title: 'Speed', value: '138.4 km/h', trend: '+1.2', trendType: 'up' },
      { title: 'Acceleration', value: '4.8 m/s²', trend: 'Steady', trendType: 'neutral' },
      { title: 'Endurance', value: '80 / 100', trend: '+1.0', trendType: 'up' },
      { title: 'Strength', value: '82 / 100', trend: '+0.5', trendType: 'up' },
```

Code Structure

The project is organized into reusable React components, making the application modular and easy to maintain.

- App Component: Controls the overall application layout and routing.
- Dashboard Component: Displays player and team performance statistics.
- Chart Components: Generate graphical representations using Recharts.

- Data Module: Stores and manages sports performance data.
- UI Components: Handle navigation, cards, buttons, and other interface elements.
- Animation Module: Uses Framer Motion to provide smooth transitions and interactive effects.

```css
src > {} index.css > ...
@import url('https://fonts.googleapis.com/css2?family=Inter:wght@300;400;500;600;700&family=Outfit:wght@400;500;600;700;800;

:root {
  /* Brand Accents */
  --victory-blue: #1565C0;
  --performance-green: #00C853;
  --energy-orange: #FF6D00;
  --championship-gold: #FFC107;
  --athletic-red: #E53935;
  --electric-cyan: #00B8D4;

  /* White & Neutral Palette */
  --bg-primary: #FFFFFF;
  --bg-secondary: #F8FAFC;
  --bg-section: #F1F5F9;
  --bg-card: #FFFFFF;
  --border-color: #E2E8F0;

  /* Text Colors */
  --text-heading: #0F172A;
  --text-body: #334155;
  --text-muted: #64748B;
  --text-disabled: #94A3B8;

  /* Sport-Specific Colors */
  --color-football: #10B981;
  --color-cricket: #2563EB;
  --color-basketball: #F97316;
  --color-volleyball: #0EA5E9;
  --color-swimming: #0284C7;
  --color-athletics: #DC2626;
  --color-badminton: #0D9488;
  --color-table-tennis: #E11D48;
  --color-cycling: #7C3AED;
  --color-gym: #475569;
  --color-boxing: #EF4444;
  --color-kabaddi: #EA580C;
  --color-tennis: #84CC16;

  /* Typography */
  --font-sans: 'Inter', sans-serif;
  --font-heading: 'Outfit', sans-serif;

  /* Premium Shadows & Radius */
  --radius-lg: 20px;
  --radius-md: 14px;
  --radius-sm: 8px;
  --shadow-sm: 0 2px 8px rgba(15, 23, 42, 0.04);
  --shadow-md: 0 8px 24px rgba(15, 23, 42, 0.05);
```

Performance Analysis

The application was evaluated based on the following criteria:

- User Interface: Responsive and easy to navigate.
- Visualization: Interactive charts accurately represent sports statistics.
- Performance: Fast page loading and smooth rendering due to React and Vite.
- Scalability: Modular architecture allows easy addition of new sports data and analytics features.
- Compatibility: Successfully tested on Google Chrome, Microsoft Edge, and Mozilla Firefox.

Conclusion of Experimentation

The experimentation demonstrates that the Sports Performance Analytics system effectively analyzes and visualizes sports performance data through an interactive dashboard. The application provides accurate statistical information, responsive visualizations, and an intuitive user experience. The results confirm that modern web technologies such as React.js, Recharts, and Vite are suitable for developing efficient and scalable sports analytics applications.

RESULT

The Sports Performance Analytics project was successfully implemented and tested, producing meaningful visual insights from sports performance data. The application effectively displays player and team statistics through an interactive dashboard, making it easier for users to analyze performance and compare different metrics. The responsive interface ensures smooth navigation across desktops and mobile devices.

The dashboard presents various performance indicators using bar charts, pie charts, and line graphs. Users can easily compare player statistics, observe team performance trends, and identify strengths and weaknesses based on historical data. The visual representation of data simplifies the interpretation of complex statistics and supports quick decision-making.

The application demonstrated fast loading speed and efficient performance due to the use of React.js and Vite. Interactive charts created using Recharts accurately represented the sports dataset, while smooth animations enhanced the overall user experience. The modular structure of the application also allows new datasets and additional analytics features to be integrated with minimal changes.

Overall, the project successfully achieved its objectives by providing an efficient, user-friendly, and visually appealing sports analytics platform. It demonstrates the importance of data visualization in sports performance analysis and provides a strong foundation for future enhancements such as real-time data integration, predictive analytics, machine learning-based performance forecasting, and personalized player recommendations.

Final Overview

The **Sports Performance Analytics** project successfully demonstrates how data analytics and modern web technologies can be used to analyze and visualize sports performance effectively. The application provides an interactive dashboard that displays player and team statistics through charts, graphs, and performance summaries, enabling users to compare performances, identify trends, and evaluate key performance metrics. Developed using **React.js**, **Vite**, and **Recharts**, the system offers a responsive, user-friendly, and efficient interface that ensures smooth performance across different devices and web browsers.

Overall, the project achieves its objective of transforming raw sports data into meaningful visual insights that support better decision-making for coaches, analysts, and sports enthusiasts. The modular design makes the application easy to maintain and extend with additional features. Future enhancements such as real-time data integration, live match analysis, machine learning-based performance prediction, and personalized recommendations can further improve the system, making it a scalable and practical solution for modern sports analytics.

CONCLUTION

The **Sports Performance Analytics** project successfully demonstrates how modern web technologies and data visualization can be used to analyze and present sports performance data in an effective and user-friendly manner. By developing an interactive dashboard, the system enables users to view player and team statistics, compare performances, and identify important trends through charts and graphs. The project simplifies the process of performance evaluation and provides meaningful insights that support better decision-making.

The application was developed using **React.js**, **Vite**, and **Recharts**, ensuring a responsive, fast, and interactive user experience. The system efficiently organizes sports data and presents it through visually appealing dashboards, making complex statistical information easy to understand. Its modular design allows for easy maintenance and future expansion while providing reliable performance across different web browsers and devices.

Overall, the project achieved its primary objective of creating a web-based sports analytics platform that transforms raw sports data into useful visual insights. It highlights the importance of data-driven analysis in improving player performance, team strategy, and overall sports management. In the future, the application can be enhanced by integrating real-time sports APIs, machine learning models for performance prediction, live match analytics, and personalized recommendations, making it an even more powerful tool for coaches, analysts, and sports enthusiasts.

REFRENCES

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